

HICA-S, Track T4 (addendum): An FD-Completion Label Rule for Algorithm A

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Abstract

The companion note (*The Local Pruning Frontier*, hereafter [LPF]) characterises the smallest pool $\mathcal{K}_i^*$ that a bid-independent, driver-local rule may retain, but computes it only *after* Algorithm A has enumerated every route. This addendum moves part of that reasoning *inside* the label-extension procedure. When a partial route has served, or picked up, a set T of orders, we bound from below — uniformly over all feasible completions and all admissible bids — how much the route would save by dropping T and sending it to the fixed fleet; if the saving is at least the FD price $q(T)$, the label and its whole subtree are discarded. We prove that the rule never discards a route of $\mathcal{K}_i^*$, so that Algorithm A with the rule followed by Π^* produces exactly the same pool, allocation, and VCG payments as before. The rule is an adaptation of *rollback pruning* (Lozano, Duque & Medaglia, 2016); the adaptation is not a formality: because working time enters the objective and waiting is allowed, the direct transfer of rollback is unsafe (Example 9), and a correction for waiting absorption (Lemma 3) is required. We also report a negative result: a forward lower-bound matrix, the most natural first idea, is too weak to ever fire on the test instances, for a structural reason. An independent brute-force audit found no violation of any bound, and mutation tests confirm that each component of the bound is necessary.

1 Setting and notation

Notation follows [LPF]. Fix a driver i with start point v_0 , departure time t_0 and coefficient κ . A *label* $L = (v_1, \dots, v_m)$ is a precedence-feasible stop sequence (pickups p_o , deliveries d_o). Travel distances $d(\cdot, \cdot)$ and times $\tau(\cdot, \cdot)$ satisfy the triangle inequality, service times $s_v \geq 0$, and a pickup p_o has an opening (release) time e_{p_o} that is met by waiting; deliveries have no opening time (Remark 8 treats the general case). Service starts and ready times are

$$a_k = \max\{e_{v_k}, r_{k-1} + \tau(v_{k-1}, v_k)\}, \quad r_k = a_k + s_{v_k}, \quad r_0 = t_0.$$

For a gigworker, $K = \kappa \cdot (\text{route length})$ and $W = (r_{\text{last}} - t_0)/60$; for an occasional driver the route ends with a leg to the personal destination (no opening time), $K = \kappa \cdot (\text{length} - \text{direct length})$ and $W = (\text{arrival at destination} - t_0 - \text{direct time})/60$. The reported cost is $c(b) = K + bW$ with $b \in \Theta = [\underline{\theta}, \bar{\theta}]$, $\underline{\theta} \geq 0$.

Definition 1 (Shortcut of a label). Let $P(L)$ be the set of orders picked up in L and $\emptyset \neq T \subseteq P(L)$. $L \setminus T$ is the subsequence of L without the stops of T ; v' is its last point ($v' = v_0$ if it is empty), r' its ready time at v' ($r' = t_0$ if empty), and D' its length. Write $D = D(L)$, v_m and $r = r_m$ for the label itself. Let $\mathcal{F}(L)$ be the set of pickups p_w of orders $w \notin P(L)$ with $r + \tau(v_m, p_w) \leq l_{p_w}$, taken

empty when $|P(L)| = B$. Define

$$\begin{aligned}\Delta D(L, T) &:= D - D' - d(v', v_m), & \delta(L, T) &:= r - r' - \tau(v', v_m), \\ A(L, T) &:= \max\left\{0, \max_{p_w \in \mathcal{F}(L)} (e_{p_w} - r' - \tau(v', p_w))\right\}, & \beta(L, T) &:= \kappa \Delta D + \underline{\theta} \frac{\max\{0, \delta - A\}}{60}.\end{aligned}$$

ΔD and δ are the distance and time that the label spends on T , measured on the part of the route that is already known; the terms $d(v', v_m)$ and $\tau(v', v_m)$ correct for the fact that the continuation may start from the last stop of L (this matters when that stop belongs to T). A bounds how much of the time saving later waiting can absorb.

The rule. *Discard the label L and every extension of it if $\beta(L, T) \geq q(T)$ for some $\emptyset \neq T \subseteq P(L)$.*

2 Two lemmas on paths

Lemma 2 (Removing stops never hurts). *Let σ' be a subsequence of a stop sequence σ , both started at (v_0, t_0) . Then the length of σ' is at most that of σ , every retained stop has a service start in σ' no later than in σ , and the completion time (and, for an occasional driver, the arrival time at the destination) does not increase.*

Proof. This is the argument of [LPF, Lemma 3], which uses only the triangle inequality, nonnegative service and waiting, and the recursion for a_k ; feasibility of σ' is not needed. $\square$

The next lemma is the new ingredient. It quantifies how a time advantage survives a common continuation when waiting is allowed.

Lemma 3 (Absorption of a time advantage). *Two paths continue with the same stops $w_1, \dots, w_k$. Path 1 leaves point x at ready time η , path 2 leaves x' at ready time η' . Let $T_1 = \eta' + \tau(x', w_1)$ and $T_{j+1} = T_j + s_{w_j} + \tau(w_j, w_{j+1})$ be the no-wait arrival times of path 2, and suppose $\eta + \tau(x, w_1) \geq T_1 + \delta$. Then for every j*

$$a_j^{(1)} - a_j^{(2)} \geq \delta - \max_{l \leq j} (e_{w_l} - T_l)^+,$$

where $a_j^{(1)}, a_j^{(2)}$ are the service starts at w_j , and $e_w = -\infty$ for stops without opening time.

Proof. Write $\text{dur}(l \rightarrow j) := \sum_{h=l}^{j-1} (s_{w_h} + \tau(w_h, w_{h+1}))$. Unrolling the recursion gives, for any path entering w_1 at no-wait arrival U_1 , $a_j = \max\{U_1 + \text{dur}(1 \rightarrow j), M_j\}$ with $M_j := \max_{l \leq j} \{e_{w_l} + \text{dur}(l \rightarrow j)\}$, and M_j is the same for both paths. For path 2, $U_1 = T_1$ and $T_1 + \text{dur}(1 \rightarrow j) = T_j$; for path 1, $U_1 \geq T_1 + \delta$. Hence $a_j^{(1)} \geq \max\{T_j + \delta, M_j\}$ and $a_j^{(2)} = \max\{T_j, M_j\}$. If $M_j \leq T_j$ the difference is at least δ ; if $T_j < M_j \leq T_j + \delta$ it is at least $T_j + \delta - M_j$; otherwise it is at least 0. In all cases it is at least $\delta - (M_j - T_j)^+$, and $M_j - T_j = \max_{l \leq j} (e_{w_l} - T_l)$ because $T_j = T_l + \text{dur}(l \rightarrow j)$. $\square$

3 Main results

Theorem 4 (Label bound). *Let L be a label, $\emptyset \neq T \subseteq P(L)$, and ρ any feasible route of driver i with prefix L (possibly $\rho = L$). Let $\rho \setminus T$ be ρ without the stops of T . Then*

$$K_\rho - K_{\rho \setminus T} \geq \kappa \Delta D(L, T), \quad W_\rho - W_{\rho \setminus T} \geq \frac{\max\{0, \delta(L, T) - A(L, T)\}}{60},$$

and therefore $c_\rho(b) - c_{\rho \setminus T}(b) \geq \beta(L, T)$ for every $b \geq \underline{\theta}$.

Proof. Write $\rho = L \cdot \sigma$ and consider the intermediate path $\pi := (L \setminus T) \cdot \sigma$. Deleting from π the stops of T that lie in σ (deliveries of orders of T still on board at the end of L) gives $\rho \setminus T$, so by Lemma 2 $K_\pi \geq K_{\rho \setminus T}$ and $W_\pi \geq W_{\rho \setminus T}$. It is therefore enough to compare ρ with π ; also $K_\rho \geq K_\pi$ and $W_\rho \geq W_\pi$ by the same lemma.

Distance. If σ is empty (a gigworker route ending at v_m), $D(\rho) - D(\pi) = D - D' \geq \Delta D$. Otherwise let u be the first point of σ (for an occasional driver possibly the destination). Then $D(\rho) - D(\pi) = D + d(v_m, u) - D' - d(v', u) \geq D - D' - d(v', v_m)$ by $d(v', u) \leq d(v', v_m) + d(v_m, u)$. For an occasional driver the direct length cancels, and both detours are nonnegative, so the positive part in K does not interfere.

Time. If σ is empty, $W_\rho - W_\pi = (r - r')/60 \geq \delta/60$ since $\tau \geq 0$. Otherwise apply Lemma 3 with $x = v_m$, $\eta = r$, $x' = v'$, $\eta' = r'$: the hypothesis holds because $r + \tau(v_m, w_1) \geq r' + \tau(v', w_1) + \delta$ follows from $\tau(v', w_1) \leq \tau(v', v_m) + \tau(v_m, w_1)$. At the last stop (and at the destination), the completion times therefore differ by at least $\delta - \max_l (e_{w_l} - T_l)^+$. A stop w_l of σ with an opening time is a pickup of an order not picked in L ; it can occur only if $|P(L)| < B$, and feasibility of ρ requires $r + \tau(v_m, w_l) \leq l_{w_l}$ (triangle inequality), so $w_l \in \mathcal{F}(L)$. Since $T_l \geq r' + \tau(v', w_l)$ (triangle inequality, nonnegative service), the maximum is at most $A(L, T)$. Combined with $W_\rho \geq W_\pi$ this gives the time bound.

Cost. $c_\rho(b) - c_{\rho \setminus T}(b) = (K_\rho - K_{\rho \setminus T}) + b(W_\rho - W_{\rho \setminus T})$ with a nonnegative slope, so it is at least its value at $\underline{\theta}$, which is at least β . $\square$

Theorem 5 (The rule preserves the local frontier exactly). *Let R_i be the pool of Algorithm A (complete enumeration, per-bundle Pareto filtering, one representative per pair (K, W)), and R'_i the pool obtained when Algorithm A in addition discards some or all labels satisfying the rule, together with their extensions. Under the assumptions of [LPF, Lemma 3] and $\underline{\theta} < \bar{\theta}$,*

$$\mathcal{K}_i^*(R'_i) = \mathcal{K}_i^*(R_i).$$

Consequently the pipeline “Algorithm A with the rule, then Π^ ” outputs exactly the same pool as before, and [LPF, Theorem 4 and Corollary 6] apply verbatim: the WDP value, every removal value and the Clarke-pivot payments are unchanged.*

Proof. Let F be the set of all feasible routes, so R_i is the Pareto representative set of F and R'_i that of F minus the discarded routes.

(i) *Discarded routes of R_i lie outside $\mathcal{K}_i^*(R_i)$.* Let $\rho \in R_i$ extend a discarded label with witness T . By Theorem 4, $c_\rho(b) \geq c_{\rho \setminus T}(b) + \beta \geq c_{\rho \setminus T}(b) + q(T)$ for all $b \in \Theta$. The route $\rho \setminus T$ is feasible ([LPF, Lemma 3]) and by downward closure there is, for every b , some $\hat{\rho} \in R_i \cup \{\mathbf{0}_i\}$ with bundle $S_\rho \setminus T$ and $c_{\hat{\rho}}(b) \leq c_{\rho \setminus T}(b)$. Since $\hat{\rho} \in D(\rho)$, $E_\rho(b) \leq c_{\hat{\rho}}(b) + q(T) \leq c_\rho(b)$; hence $\mu_\rho \leq 0$ on Θ and $\rho \notin \mathcal{K}_i^*(R_i)$.

(ii) $\mathcal{K}_i^*(R_i) \subseteq R'_i$. A route of $\mathcal{K}_i^*(R_i)$ is not discarded by (i), and it is Pareto-undominated in F , hence in any subset of F containing it.

(iii) $\mathcal{K}_i^*(R_i) \subseteq \mathcal{K}_i^*(R'_i)$. Take $r \in \mathcal{K}_i^*(R_i)$. Compared with $D_{R_i}(r)$, the set $D_{R'_i}(r)$ lacks some routes (which can only raise E_r) and may contain routes r_2 that were Pareto-dominated in F ; such an r_2 is dominated by a route $r_1 \in R_i$ with the same bundle, so $c_{r_2} + q(\cdot) \geq c_{r_1} + q(\cdot)$, which is at least $E_r^{R_i}$ when $r_1 \neq r$; when $r_1 = r$, Pareto dominance gives $c_{r_2}(b) > c_r(b)$ for every $b > 0$. Hence on the nonempty open set where $\mu_r^{R_i} > 0$ (minus possibly $b = 0$), $E_r^{R'_i} > c_r$, and $r \in \mathcal{K}_i^*(R'_i)$.

(iv) $\mathcal{K}_i^*(R'_i) \subseteq \mathcal{K}_i^*(R_i)$. Let $r \in R'_i$ with $r \notin \mathcal{K}_i^*(R_i)$. If $r \in R_i$, [LPF, Lemma 5] gives, for each b , some $u \in \mathcal{K}_i^*(R_i) \cup \{\mathbf{0}_i\}$ with $S_u \subseteq S_r$ and $c_u(b) + q(S_r \setminus S_u) \leq c_r(b)$; $u \in R'_i$ by (ii), so $\mu_r^{R'_i} \leq 0$. If $r \notin R_i$, r is Pareto-dominated in F by some $r_1 \in R_i$ with the same bundle; r_1 must have been discarded (otherwise r would not survive the Pareto filter of R'_i), so $r_1 \notin \mathcal{K}_i^*(R_i)$ by (i),

and the same lemma applied to r_1 yields $u \in R'_i$ with $c_u + q(S_r \setminus S_u) \leq c_{r_1} \leq c_r$. In both cases $r \notin \mathcal{K}_i^*(R'_i)$. $\square$

Remark 6 (What the rule is and is not). The rule reads no bid (only the public constant $\underline{\theta}$ and FD prices) and uses only driver i 's own data, so it is a local, bid-independent rule in the sense of [LPF]. By [LPF, Theorem 4(c)], *no* rule of this kind can discard a route of $\mathcal{K}_i^*$; the best any such label rule can do is to cut every label whose subtree contains no route of $\mathcal{K}_i^*$. Section 6 measures how close the rule comes to that ceiling.

Remark 7 (Combination with label dominance). If Algorithm A also deletes a label L_2 because another label L_1 with the same end stop and the same picked/delivered sets dominates it (ready time, length and cost no larger), every completion of L_2 is dominated by the corresponding completion of L_1 : the recursion for a_k is nondecreasing in the starting time, so the same continuation from L_1 is feasible, no longer and finishes no later. If L_1 is discarded by the rule, its completions lie outside $\mathcal{K}_i^*$, and so do those of L_2 , since a same-bundle route that costs no less for every b cannot be in $\mathcal{K}_i^*$. The proof of Theorem 5 is unchanged.

Remark 8 (Delivery time windows with an opening time). If deliveries also have opening times, include in $\mathcal{F}(L)$ the deliveries of orders on board at the end of L and of every order that may still be picked up. Lemma 3 is unchanged.

4 Why each term is needed: a counterexample and a negative result

Example 9 (Direct transfer of rollback pruning is unsafe). Rollback pruning compares a path with the same path skipping a node and prunes if skipping is not worse; in the ESPPRC this is safe because costs are additive over arcs and time is only a resource, so arriving earlier never costs anything. Here working time is part of the cost and waiting is allowed. Transferring rollback directly amounts to setting $A \equiv 0$ in Definition 1. On a synthetic instance (seed 2, gigworker g_0 , $B = 3$), for the label $L = (p_{o_2}, d_{o_2})$ and $T = \{o_2\}$ with $q_{o_2} = 17.13$, the uncorrected bound credits a time saving of 42.9 min, giving $17.72 \geq q_{o_2}$, and the label is discarded. In the feasible completion $(p_{o_2}, d_{o_2}, p_{o_0}, d_{o_0})$, however, the pickup of o_0 opens late: without o_2 the driver would arrive early and wait, so the true time saving is only 7.8 min and the true saving at $b = \underline{\theta}$ is $7.24 < 17.13$. Serving o_2 in this route is therefore worth more than delegating it to FD, and the uncorrected rule deletes a route it must keep. With the correction A the label is kept.

The mutation tests in Section 6 show the same at scale for both correction terms.

Negative result: a forward lower-bound matrix does not work. The most natural first idea is to decide *before* visiting p_o : from the current stop v , bound the extra distance by $\min_x \{d(v, p_o) + d(p_o, x) - d(v, x)\}$ over all possible next stops x (a matrix that can be precomputed), and the extra time analogously. This rule is safe (the same lemmas apply) but on the synthetic instances it never fired. The median distance credit was 0.5–0.75 USD against FD prices of 15–21 USD, because some other stop almost always lies close to the direction of p_o ; and the median time credit was 0, because the detour (about 5 min) is fully absorbed by the waiting for later-released pickups (median absorption 34–46 min). Both reasons are structural: before p_o is visited, the continuation is unknown, and the bound must hold for the cheapest one. The rule of this note is the same idea evaluated one stop *after* p_o , when the detour to p_o is known exactly; the correction terms are what make that evaluation safe for every continuation.

5 Implementation

Each label carries, for every picked order j , the triple (D'_j, v'_j, r'_j) of its shortcut $L \setminus \{j\}$, as an additional resource. Extending the label by a stop u updates each triple in $O(1)$: if u is a stop of j the triple is unchanged; otherwise the shortcut travels to u exactly like the label. When j is picked up, its triple is the parent label's state. The check first evaluates β with $A = 0$, which is an upper bound; only if that reaches q_j is A computed. The per-extension overhead is therefore $O(|P(L)|) \leq O(B)$ plus rare $O(n)$ evaluations of A . Subsets T with $|T| \geq 2$ added almost nothing in the experiments and are not needed.

6 Evidence

All numbers come from the independent brute-force reference implementation of [LPF] (synthetic Euclidean instances, $n = 8$ orders, two gigworkers and two occasional drivers, $\Theta = [18, 25]$). They check the mathematics; they are not thesis results.

Check	$B = 3$ (3 seeds)	$B = 4$ (2 seeds)
Rule firings audited	5,868	33,292
Feasible completions checked against the bound	9,975	87,744
Bound violations	0	0
$\mathcal{K}^*$ identical with and without the rule	yes	yes
Max WDP value difference (random bids)	2.8×10^{-14}	1.3×10^{-13}
Mutation $A \equiv 0$ (direct rollback transfer)	1,339 violations, $\mathcal{K}^*$ changed	—
Mutation without junction terms $d(v', v_m), \tau(v', v_m)$	4,173 violations, $\mathcal{K}^*$ changed	—

In both mutations the WDP values on 40 random bid profiles per seed were still identical: the wrongly deleted routes are rarely optimal, so bid sampling alone would not detect the error. The completion-level bound audit and the $\mathcal{K}^*$ comparison did.

Distance to the ceiling. Counting labels that lie on a path to some feasible route: an ideal local rule (Theorem 4(c) of [LPF]) keeps 2.7% of them at $B = 3$ and 0.6% at $B = 4$; the rule keeps 56% and 50%, i.e. it closes 45% and 50% of the achievable gap.

Work and time. With the incremental implementation, label extensions fell by 19–44% at $B = 3$ and 26–61% at $B = 4$. Wall-clock time in this plain enumerator ranged from $0.8\times$ to $2\times$ the baseline. The decisive factor is the FD price level relative to route costs (median over three seeds and two gigworkers, $B = 4$):

FD prices scaled by	0.6	1.0	1.4
Label extensions saved	73–76%	27–30%	7–9%
Wall-clock speed-up	$2.8\text{--}3.1\times$	$\approx 1.0\times$	$0.75\times$

Compressing release times by a factor of four changed these figures by at most three points. On the RQ1 instances Π^* removes 77–97% of the pool, against about 61% on these synthetic instances, which suggests that FD is relatively cheaper there and that the rule is in its effective regime; this must be measured, not assumed.

7 Position in the literature

The principle — compare a partial path with itself minus a node and prune if the node is not worth its price — is the rollback pruning of Lozano, Duque & Medaglia (2016, *Transportation Science*, §4.3), built on the dominance relation of Feillet et al. (2004); there the price is the node’s dual value. It must be cited as the origin of the rule. The adaptation adds four things, each needed for correctness or for the mechanism: (a) the correction A for waiting absorption, without which the rule is unsafe when time enters the objective (Example 9); (b) removal of pickup–delivery *pairs*, including orders still on board and a removed last stop (junction terms); (c) comparison uniformly over Θ with a bid-independent price, so that the pool stays fixed before bids; (d) the exact guarantee of Theorem 5 that the local frontier, and hence the VCG outcome, is unchanged. Label-dominance rules for the PDPTW (Dumas, Desrosiers & Soumis, 1991; Ropke & Cordeau, 2009) and bounding strategies of the pulse family have not yet been checked for a closer precedent; this must be done before (a)–(d) are claimed as new.

References to verify before use

Lozano, L., Duque, D. & Medaglia, A.L. (2016), An exact algorithm for the elementary shortest path problem with resource constraints, *Transportation Science* (rollback pruning, §4.3; read directly from the project copy, online version dated 2015). Feillet, D., Dejax, P., Gendreau, M. & Gueguen, C. (2004), *Networks* (dominance for the ESPPRC; not read directly). Dumas, Desrosiers & Soumis (1991), *EJOR*. Ropke & Cordeau (2009), *Transportation Science*.